Supplementary Material for "Confidence-Gated Dual-Process Fall Detection: Retrieval-Augmented Reasoning and Cost-Sensitive Response on Wearable Inertial Sensor Data"

Table and section numbers prefixed with "S" refer to this document; all other numbers refer to the main paper.

S-I. Reproducibility Notes

The baseline mode is called `primary_only` in the paper; released configurations and result files use the legacy key `tier1_only` (`configs/tier1_sisfall.yaml`). Main classification results use the supplied main-ablation confusion counts, one observation per mode and fold; latency and escalation come from a separate timing run with identical TP/FN counts and FP differences of at most 2 windows per fold. The shipped `configs/paper_protocol.yaml` sets `primary_backbone: cnn_lstm_attn` to match the main CNN–LSTM–Attention tables; HARMamba is a generality backbone only. The following scripts reproduce the tabulated arithmetic without a LaTeX installation:

```
python check_manuscript.py   # inputs, labels, cites, arity
python check_numbers.py      # Mean rows vs. per-fold rows
python check_reported_results.py # counts vs. tables
python recompute_ci.py       # paired fold intervals
```

S-II. Protocol Details

a) Budgets: Seed 42; `max_train` = 40000, `max_val` = 5000, `max_cal` = 2000; main `max_test` = 1000 with a D18/D19 quota of 80 windows/code; LLM comparison 150 escalated cases/fold; feature adjudicators 160/fold; costs FN=10, FP=1.

b) Gate grid: τ_{low}: 20 points in $[0.05, 0.45]$; τ_{high}: 20 points in $[0.55, 0.95]$; objective $\text{FP} + \beta\,\text{FN} + 0.5 \cdot$ (deferred windows) over edge decisions; at least 20% edge decisions (or 10 windows) per candidate pair.

c) Reasoner: Ollama at `127.0.0.1:11434`; `mistral:latest` (primary), `qwen2.5-coder:7b-instruct` (comparison); temperature 0.1; timeout 60 s; two retries; heuristic fallback. Output JSON keys: `prediction`$\in$ {`fall`, `adl`}, `severity`$\in$ {`none`, `mild`, `moderate`, `severe`}, `confidence`$\in [0,1]$, `rationale`, `action`. Prompts are in `prompts/`.

d) Action map (simulated): Fall: impact $\geq 5\,g$ or stillness ≥ 10 s $\rightarrow$ emergency; impact $\geq 3\,g$ or stillness ≥ 3 s $\rightarrow$ notify; otherwise monitor. ADL: impact $\geq 2\,g$ or free-fall ≥ 0.12 s $\rightarrow$ monitor; otherwise log.

TABLE S1
FIVE-FOLD DETECTOR-ONLY SISFALL CONTEXT AT THE REPORTED YOUDEN OPERATING POINTS, 20,000 STANDARD-PROTOCOL WINDOWS/FOLD (PREVALENCE 0.329–0.345). E1 AND RETRAINED ZOO ARE DISTINCT CHECKPOINT RUNS. MEAN±SAMPLE STD ARE RECOMPUTED FROM THE SUPPLIED THREE-DECIMAL FOLD F1 VALUES; LAST-DIGIT DIFFERENCES FROM SEPARATELY SUPPLIED AGGREGATE SUMMARIES CAN RESULT FROM ROUNDING. NEITHER RUN IS THE FROZEN MAIN COMPARISON OR THE SEPARATE 30,000-WINDOW V2 EVALUATION. SCORES DESCRIBE TESTED IMPLEMENTATIONS AND TRAINING SETTINGS, NOT A UNIVERSAL ARCHITECTURE RANKING.

Model	E1 F1	Retrained F1
CNN1D	0.701±0.031	0.734±0.026
LSTM	0.693±0.016	0.711±0.026
CNN–LSTM	0.696±0.010	0.724±0.036
CNN–LSTM–Attention	0.717±0.022	0.739±0.030
TCN	0.702±0.029	0.747±0.034
ModernTCN	0.743±0.015	0.760±0.030
TSMixer	0.671±0.022	0.686±0.018
PatchTST	0.541±0.009	0.566±0.014
Transformer	0.690±0.013	0.703±0.017

TABLE S2
MULTI-SEED ROBUSTNESS ON SISFALL FOLD 0 (n=1000, OLLAMA/MISTRAL BACKEND). SEEDS REDRAW THE STRATIFIED TEST SLICE AS WELL AS TRAINING RANDOMNESS, SO ABSOLUTE VALUES DIFFER FROM THE FOLD 0 ROW OF THE MAIN-PAPER PER-FOLD TABLE (TABLE II, WHICH USES THE LOCKED SEED-42 SLICE); THE QUANTITY OF INTEREST IS THE ACROSS-SEED SPREAD, NOT THE LEVEL.

Seed	Primary F1	Stack F1	Stack Rec.	Cost/1k
42	0.747	0.909	0.998	95
43	0.740	0.907	0.995	105
44	0.746	0.899	1.000	96
Mean±std	0.745±0.004	**0.905±0.005**	0.998±0.003	99±6

S-III. Supplementary Results

Tables S1–S9 report detector-only context (backbone zoo, seeds, sensor perturbation), gate operating points, per-fold KFall results, cost-ratio sensitivity, rationale grounding, escalation and latency. The robustness and seed runs use different test slices from the main protocol, so their absolute values differ slightly from the per-fold table of the main paper (Table II).

TABLE S3

SENSOR-PERTURBATION ROBUSTNESS OF PRIMARY-DETECTOR ROUTING UNDER THE COST GATE (SISFALL, FIVE-FOLD MEAN F1). ROTATIONS APPLY A RANDOM RIGID ROTATION OF THE SENSOR FRAME WITH THE STATED BOUND, EMULATING GARMENT OR MOUNTING DRIFT; RESAMPLING EMULATES A DIFFERENT DEVICE SAMPLING RATE. DEGRADATION IS GRACEFUL—THE WORST CASE, $\pm 15^\circ$, LOSES 0.014 F1—SO THE AMBIGUOUS BAND IS NOT AN ARTEFACT OF ONE EXACT SENSOR PLACEMENT.

Condition	Mean F1	Δ vs. clean
clean	0.768 ± 0.017	—
resample 100 Hz	0.766 ± 0.016	-0.002
resample 50 Hz	0.766 ± 0.015	-0.002
rotate $\pm 10^\circ$	0.763 ± 0.020	-0.005
rotate $\pm 15^\circ$	0.754 ± 0.023	-0.014

TABLE S4

ARCHIVED POOLED ROUTING DIAGNOSTIC WITHOUT SYSTEM 2. THE ENTRIES LABELLED MIN-COST AND Q25 TIE AT REPORTED COST 1015, WHILE Q25 HAS 23 PERCENTAGE POINTS LESS DEFERRAL. UNDER THESE DISPLAYED OBJECTIVES, MIN-COST IS DOMINATED; THE SOURCE SELECTION RULE AND THE TREATMENT OF DEFERRED-CASE ERRORS MUST BE DISTINGUISHED FROM THE PRODUCTION ROUTING SURROGATE. THESE POOLED POINTS DO NOT JUSTIFY CHOOSING DEPLOYMENT THRESHOLDS WITHOUT HELD-OUT CALIBRATION AND ADJUDICATOR EVALUATION.

Operating point	τ_{low}	τ_{high}	Esc. rate	Cost
Min cost	0.05	0.60	0.57	1015
Q25 esc. (dominant)	**0.30**	**0.70**	**0.34**	**1015**
Q50 esc.	0.40	0.95	0.48	1042
Q75 esc.	0.30	0.95	0.61	1223

TABLE S5

SECOND-DATASET EVALUATION ON KFALL, FROZEN PRIMARY DETECTOR VERSUS FULL STACK UNDER THE BINARY FALL/ADL PROTOCOL (n=1000/FOLD). THIS IS NOT EVIDENCE OF ZERO-SHOT TRANSFER OF SISFALL-TRAINED WEIGHTS OR FORMAL NON-INFERIORITY. FOLD 1 REGRESSES; FOLD 4 HAS THE LARGEST DISPLAYED F1 GAIN. DELTAS BELOW ARE COMPUTED FROM THE DISPLAYED ROUNDED VALUES. FOLD 4 PRIMARY ROW IS CONSISTENT WITH ITS CONFUSION COUNTS (TP=473, TN=508, FP=9, FN=10; 483 FALLS, NOT ≈499 AS IN OTHER FOLDS): F1 0.980, RECALL 0.979, COST 109.

Fold	n	`primary_only`			`gate_knn_llm`			ΔF1	ΔCost
		F1	Rec.	Cost/1k	F1	Rec.	Cost/1k		
0	1000	0.992	0.998	17	0.993	1.000	7	+0.001	−10
1	1000	0.987	0.996	31	0.986	0.996	32	−0.001	+1
2	1000	0.987	0.994	40	0.989	0.996	29	+0.002	−11
3	1000	0.992	0.990	53	0.995	0.996	23	+0.003	−30
4	1000	0.980	0.979	109	0.989	0.994	38	+0.009	−71
Mean	1000	0.987	0.991	50	**0.990**	**0.996**	**26**	+0.003	−24.2

TABLE S6

SENSITIVITY TO THE CLINICAL COST RATIO $\beta = c_{\mathrm{FN}}/c_{\mathrm{FP}}$ (SISFALL, FIVE-FOLD MEAN COST PER 1000 WINDOWS, $\mathcal{R}_\beta = \beta \cdot \mathrm{FN} + \mathrm{FP}$). THE ADVANTAGE OF THE STACK IS MONOTONE IN β: THE MORE A DEPLOYMENT PENALISES MISSED FALLS RELATIVE TO NUISANCE ALARMS, THE LARGER THE ABSOLUTE SAVING. $\tau^* = 1/(\beta{+}1)$ IS THE CORRESPONDING HARD-DECISION BAYES THRESHOLD (**??**). THE β=10 ROW IS THE PAPER'S PRIMARY SETTING.

β	τ^*	Mode	Cost/1k	Δ
5	0.167	`primary_only`	628	—
5	0.167	**`gate_knn_llm`**	**149**	−479
10	0.091	`primary_only`	1151	—
10	0.091	**`gate_knn_llm`**	**200**	−951
20	0.048	`primary_only`	2197	—
20	0.048	**`gate_knn_llm`**	**302**	−1895

TABLE S7

RATIONALE GROUNDING. $\mathcal{G}_{\mathrm{faith}}$ IS THE FRACTION OF RATIONALES THAT MENTION AT LEAST ONE EVIDENCE FIELD PRESENT IN $\Sigma(\mathbf{x})$. IT IS A PROMPT-COMPLIANCE CHECK, NOT A MEASURE OF REASONING FAITHFULNESS OR CORRECTNESS: A RATIONALE CAN CITE A FIELD AND STILL REACH THE WRONG CONCLUSION. *Graded cost* SCORES THE FOUR-LEVEL SIMULATED ACTION RATHER THAN THE BINARY LABEL. CONTRASTIVE CRITIQUE WITH FROZEN LABELS RAISES $\mathcal{G}_{\mathrm{faith}}$ (0.436→0.843) AND LOWERS GRADED ACTION COST BY 45%.

Fold	$\mathcal{G}_{\mathrm{faith}}$		Graded action cost	
	base	+critique	base	+critique
0	0.430	0.851	544	268
1	0.340	0.749	901	599
2	0.471	0.871	535	261
3	0.482	0.874	428	216
4	0.459	0.868	537	274
Mean	0.436	**0.843**	589	**324**

TABLE S8

SYSTEM 2 DUTY CYCLE BY EVALUATION PROTOCOL. THE 1000-WINDOW SISFALL RATE COMES FROM THE SEPARATE TIMING RUN, NOT THE PRIMARY CLASSIFICATION COUNTS. SISFALL PROTOCOLS ARE STRATIFIED AND D18/D19 ENRICHED: THESE RATES CONSTRAIN FEASIBILITY BUT ARE NEITHER FREE-LIVING ESTIMATES NOR GUARANTEED UPPER BOUNDS. DISTRIBUTION SHIFT CAN CHANGE ESCALATION IN EITHER DIRECTION. BINARY KFALL INVOKES DELIBERATION ON ROUGHLY ONE WINDOW IN NINETY UNDER ITS OWN PROTOCOL.

Protocol	n/fold	Escalation rate
SisFall timing (5-fold mean)	1000	0.47 ± 0.08
SisFall ambiguous stress	300	0.59 ± 0.04
KFall external (5-fold mean)	1000	0.011

TABLE S9

PROCESSING LATENCY FROM THE SEPARATE OLLAMA/MISTRAL AGENTIC TIMING RUN (SISFALL, n=1000/FOLD; GPU EVALUATION, NOT WEARABLE HARDWARE). THE PRIMARY DETECTOR IS EVALUATED FOR EVERY WINDOW; THE ESCALATED PATH IS DOMINATED BY LOCAL LLM DECODING. MEAN STACK IS THE MEASURED PER-WINDOW AVERAGE. FOLD 2 HAS THE SLOWEST ESCALATED PATH (22.85 S), NOT THE HIGHEST DUTY CYCLE (ITS RATE IS 0.460, VERSUS 0.546 ON FOLD 4). WINDOW ACQUISITION, QUEUEING AND CLINICAL RESPONSE TIME ARE NOT ESTABLISHED BY THESE MEASUREMENTS.

Fold	Primary (ms)	Escalated (s)	Mean stack (s)
0	8.96	12.81	5.68
1	8.09	10.66	3.73
2	8.52	22.85	10.52
3	8.36	11.00	5.99
4	8.89	10.86	5.93
Mean	8.56	13.64	6.37